ASIANZU BENJAMIN CHRIS

Health Data Analyst • Machine Learning • Public Health Informatics

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PROFESSIONAL SUMMARY

Health Data Analyst and Machine Learning practitioner with demonstrated experience delivering predictive analytics systems in real-world public health settings across East Africa. Built and deployed a disease outbreak risk prediction system for a regional public health NGO, achieving 84% model accuracy and an estimated 28% reduction in malaria cases through early intervention. Co-developed a live health records system now deployed at Maracha Hospital, West Nile Uganda. Combines deep technical capability in Python, SQL, and machine learning with domain knowledge in infectious disease surveillance, healthcare informatics, and rural health systems. Open to remote roles with WHO, UNICEF, international NGOs, and global health analytics organizations.

CORE COMPETENCIES

Machine Learning: scikit-learn, random forest, gradient boosting, logistic regression, ROC-AUC, recall optimization

Health Informatics: disease surveillance, outbreak detection, health records systems, rural health data infrastructure

Data Engineering: ETL pipelines, multi-source data integration, lag variable engineering, interpolation, SQLAlchemy

Languages & Tools: Python (pandas, numpy, scikit-learn, matplotlib), R (ggplot2, tidyverse), SQL (PostgreSQL, MySQL)

Visualization: Power BI (DAX, heatmaps, risk dashboards, KPI forecasting), ggplot2, matplotlib

Data Sources: NASA climate datasets, World Bank demographic indicators, district health surveillance reports

Soft Skills: Stakeholder reporting, executive dashboards, curriculum design, NDA client delivery

PROFESSIONAL EXPERIENCE

Data Analyst & IT Officer | Muni Labs, West Nile, Uganda *2023 – Present*

- Co-developed and deployed a Health Records System at St. Joseph's Hospital, Maracha — a live digital infrastructure project transforming patient data management in a rural healthcare setting.
- Conducted infrastructure gap analysis and issued technical recommendations (server specs, LAN architecture) enabling real-time multi-user system deployment.
- Designed and delivered a 12-week Introduction to Data Analysis course (Excel + R) for the Graduate Studies Academy, covering EDA, statistical modeling, visualization, and capstone projects.
- Provided ongoing data analysis, IT support, and software solutions across NGO health and research operations.

Freelance Health Data Analyst | Regional Public Health NGO, East Africa (NDA) *2024*

- Contracted to build a predictive analytics system for early disease outbreak detection across 45 districts in East Africa.
- Integrated 7 years of multi-source surveillance data (180,000+ observations) from district health offices, NASA climate datasets, and World Bank demographic indicators.

- Engineered lag features linking rainfall patterns to malaria cases 4–6 weeks later. Key correlations: rainfall vs malaria 0.62, temperature anomaly vs outbreak intensity 0.48.
- Trained and evaluated logistic regression, random forest, and gradient boosting classifiers. Final model: Random Forest with 84% accuracy, ROC-AUC 0.89, recall 0.81.
- Delivered interactive Power BI dashboard with national risk heatmap, 30-day outbreak probability forecasting, and intervention prioritization panel for field response teams.
- Simulation models estimated 28% potential reduction in malaria cases when early intervention strategies were deployed based on system alerts.

Course Instructor – Python & Programming | ReedInc Org, West Nile, Uganda *Feb 2024*

- Developed and delivered beginner Python programming curriculum covering fundamentals, algorithmic thinking, and applied problem solving for community learners.

Biochemistry Teacher | Secondary School, Uganda *2023 – Present*

- Teach O-level and A-level Biology and Chemistry. Developed lesson plans and assessments aligned to Uganda National Curriculum.

KEY PROJECTS

Disease Outbreak Risk Prediction System — *Paid NDA Client Project*

- End-to-end ML pipeline: data integration → feature engineering → model training → Power BI risk dashboard → field deployment.
- Random Forest classifier: 84% accuracy, ROC-AUC 0.89, recall 0.81. Top predictors: 4-week rainfall lag, temperature anomaly, sanitation coverage, population density.
- System flagged districts exceeding 70% outbreak probability during high rainfall periods, enabling proactive resource deployment.

Maracha Hospital Health Records System — *Live Deployment*

- Co-developed digital health records infrastructure for St. Joseph's Hospital, Maracha — a rural hospital in West Nile, Uganda.
- System now in active testing and deployment, improving patient record management and reducing administrative burden for clinical staff.

E-Commerce Revenue Intelligence System

- Full analytics pipeline on 98,666 orders across 3 years. Identified 87.7% churn rate with R$1.38M recoverable revenue. Delivered Python → PostgreSQL → Power BI → executive board presentation.

EDUCATION

Bachelor of Science in Education (Biology & Chemistry) | Muni University, Arua, Uganda *Feb 2023*

Uganda Advanced Certificate of Education | Our Lady of Africa SS, Mukono *Dec 2017*

CERTIFICATIONS

- Microsoft Certified: Data Analyst Associate (Power BI)
- Career Skills in Data Analytics — LinkedIn Learning
- Data Analysis with Python — freeCodeCamp
- Certified Python Programmer — DataCamp, 2024
- Certified R Programmer — R Foundation, 2022